

SET-6(DSA0611-DH&V)

Product Sales Analysis

AIM

To analyze monthly product sales data and visualize the sales of different products using grouped bar charts, stacked area charts, tables, and a Tableau dashboard.

DATASET

Product ID	Product Name	January Sales	February Sales	March Sales
1	Product A	2000	2200	2400
2	Product B	1500	1800	1600
3	Product C	1200	1400	1100

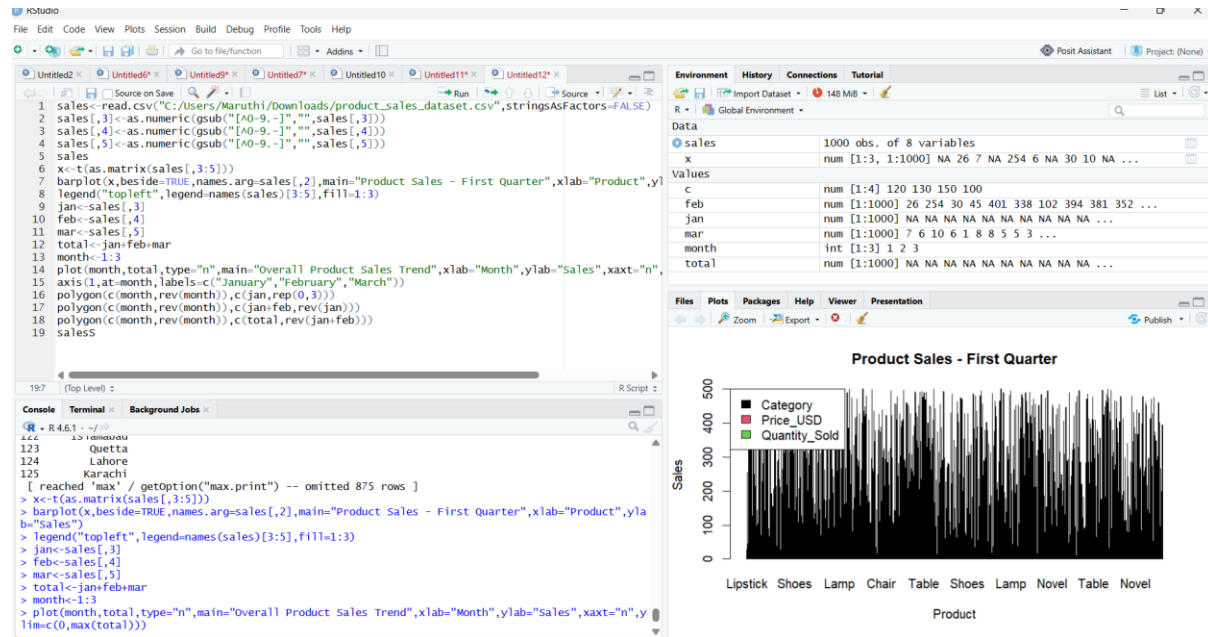
ALGORITHM

1. Start Tableau.
2. Create the Product Sales dataset with Product ID, Product Name, and monthly sales.
3. Connect the dataset to Tableau.
4. Create a grouped bar chart using Product Name and monthly sales.
5. Add suitable title and axis labels.
6. Create a stacked area chart to visualize the sales trend of all products.
7. Create a table showing the monthly sales of each product.
8. Create a new Tableau dashboard.
9. Add the grouped bar chart, stacked area chart, and table to the dashboard.
10. Enable interactive filtering and tooltips.
11. Display the dashboard.
12. Stop.

R CODE

```
sales<-data.frame(Product=c("Product A","Product B","Product C"),January=c(2000,1500,1200),February=c(2200,1800,1400),March=c(2400,1600,1100))
barplot(t(as.matrix(sales[,2:4])),beside=TRUE,names.arg=sales$Product,main="Product Sales - First Quarter",xlab="Product",ylab="Sales",col=1:3)
legend("topleft",legend=names(sales)[2:4],fill=1:3)
x<-1:3
plot(x,sales$January,type="n",main="Overall Product Sales Trend",xlab="Month",ylab="Sales",xaxt="n",ylim=c(0,7000))
axis(1,x,labels=c("January","February","March"))
polygon(c(x,rev(x)),c(sales$January,rep(0,3)))
polygon(c(x,rev(x)),c(sales$January+sales$February,rev(sales$January)))
polygon(c(x,rev(x)),c(sales$January+sales$February+sales$March,rev(sales$January+sales$February)))
sales
```

OUTPUT



RESULT

Thus, the Product Sales data was successfully analyzed using Tableau. The grouped bar chart, stacked area chart, sales table, and interactive dashboard were successfully created to visualize first-quarter product sales.